Tutorial Sheet 6

IIT Delhi – Abu Dhabi

Course Code: AENL226

Course Name: Power Electronics and Power Systems

Buck and Boost Converters

1. An ideal buck converter has $V_{\text{in}} = 60\,\text{V}$, $V_o = 24\,\text{V}$, $L = 240\,\mu\text{H}$, $f_s = 20\,\text{kHz}$, and load resistance $R = 12\,\Omega$.

   Determine: (a) the duty ratio, (b) the inductor-current ripple, (c) the critical load resistance value when $I_{L,\text{min}} = 0$, and (d) the maximum switch current and maximum switch blocking voltage.

2. An ideal boost converter operates in CCM with $V_{\text{in}} = 30\,\text{V}$, $V_o = 60\,\text{V}$, switching frequency $f_s = 80\,\text{kHz}$, and output power $P_o = 180\,\text{W}$.

   Select the minimum inductance so that the peak-to-peak inductor-current ripple is no more than 25% of the average input current. Also find the peak and minimum inductor currents.

3. An ideal buck converter operates in continuous conduction mode (CCM) with $V_{\text{in}} = 48\,\text{V}$, $V_o = 16\,\text{V}$, $L = 120\,\mu\text{H}$, switching frequency $f_s = 60\,\text{kHz}$, and load resistance $R = 8\,\Omega$. Neglect the equivalent series resistance (ESR) of the output capacitor.

   Determine: (a) the duty ratio, (b) the peak-to-peak inductor-current ripple, (c) the peak and minimum inductor currents, (d) the minimum output capacitance required to limit the peak-to-peak output-voltage ripple to 0.5% of V_o, and (e) the critical load resistance at the CCM/DCM boundary.

4. An ideal boost converter has $V_{\text{in}} = 18\,\text{V}$, $V_o = 54\,\text{V}$, $L = 180\,\mu\text{H}$, switching frequency $f_s = 50\,\text{kHz}$, and load resistance $R = 27\,\Omega$. Neglect the equivalent series resistance (ESR) of the output capacitor.

   Determine: (a) the duty ratio, (b) the output current, output power, and average input current, (c) the peak-to-peak inductor-current ripple, (d) the peak and minimum inductor currents and whether the converter operates in CCM, (e) the minimum output capacitance required to limit the peak-to-peak output-voltage ripple to 1% of V_o, and (f) the maximum switch current and maximum switch blocking voltage.